Speculative Borrowing and Bank Money Creation: A Calibrated DSGE Research Note on Iran

Abstract

This research note develops a closed-economy cash-in-advance dynamic stochastic general equilibrium model to study whether speculative activity can contribute to bank money creation, liquidity growth, inflation, and output weakness. The central mechanism is that higher speculative returns can make borrowers more willing to divert part of bank credit away from production and toward financial or non-financial speculation. If banks accommodate the associated demand for liquidity, speculation can affect not only credit allocation but also the quantity of money created in the economy.

The model is calibrated to the Iranian economy and includes a representative household, productive firms, a banking sector, and a consolidated government-monetary authority. The key parameter is the productive loan-allocation share, denoted by q, which governs the share of bank loans used in production. Because this share is difficult to observe directly, the model is solved under four values: $q = 0.2$, 0.4, 0.6, and 0.8. The model is driven by a technology shock and a speculative-return shock.

Impulse-response results suggest that a positive speculative-return shock increases money creation, prices, and inflation while reducing output. Under the present calibration, the responses are numerically identical for $q = 0.2$, 0.4, and 0.6: a one-percent speculative-return shock raises the model's money/liquidity measure by approximately 2.0733 percent, increases prices and inflation by 1.1345 percent, and reduces output by 0.1345 percent. For $q = 0.8$, money/liquidity rises by 2.0625 percent, prices and inflation rise by 1.5525 percent, and output falls by 0.1841 percent. These findings support the possibility of a speculative money-supply channel, while also suggesting that policies aimed at reducing non-productive speculative returns should be viewed as complements to broader monetary, fiscal, and credit-allocation measures rather than as stand-alone solutions.

Introduction and Motivation

A large literature in monetary economics studies the relationship between credit demand, bank lending, and money creation. In the endogenous-money tradition, money supply is not treated as mechanically fixed by the monetary authority alone; it may also respond to credit demand, bank behavior, and institutional features of the financial system. Less attention, however, has been given to a narrower question: can speculative activity itself contribute to money creation through the banking system?

This question matters because speculation can affect both credit allocation and aggregate liquidity. When speculative returns rise, borrowers may demand more credit for activities that do not directly expand production. If banks accommodate this demand, liquidity may grow while productive investment weakens. In this case, speculation becomes not only an asset-market issue but also a monetary and real-sector issue.

Iran provides a relevant setting for studying this mechanism. Over recent decades, the Iranian economy has experienced persistent inflation, rapid liquidity growth, and repeated speculative activity in financial and non-financial markets. Earlier descriptive evidence shows a large divergence between real output growth, liquidity growth, and price growth in Iran. For example, between 1973 and 2014, real GDP increased by 2.08 times, while liquidity increased by 15,103 times and prices increased by more than 1,000 times (Bakhshi Dastjerdi et al., 2019). This macroeconomic environment motivates the question of whether speculative borrowing can contribute to money creation and macroeconomic instability.

The existing literature has discussed the relationship between speculation and money creation mostly indirectly. Earlier work in the endogenous-money tradition examines the role of credit demand, bank lending, and institutional conditions in the money-creation process (Dow, 1986; Lavoie, 1984; Moore, 1988; Niggle, 1991). Some studies consider speculative activity in financial markets, while others analyze broader transaction and liquidity demand without restricting attention to one specific market. Fontana (2017), for example, revisits debates within endogenous-money analysis, while Howells and Beifang-Frisancho Mariscal (1992) and Howells and Hussein (1999) connect transaction activity, loan demand, and monetary dynamics in the United Kingdom.

Direct analysis of the relationship between speculative activity and bank money creation remains limited. To my knowledge, prior Persian and English-language studies have not directly examined the relationship between speculative activity and bank money creation within a DSGE framework. This research note addresses that gap by developing a closed-economy cash-in-advance DSGE model calibrated to Iran.

The model includes a representative household, productive firms, a banking sector, and a consolidated government-monetary authority. The key mechanism is introduced through the loan market: only part of bank credit is used in production, while the remaining share may be redirected toward speculative activity when speculative returns rise. This structure links speculative incentives to liquidity demand, bank money creation, inflation, and output.

The analysis focuses on three questions. First, can speculative incentives contribute to bank money creation? Second, how do shocks to speculative returns affect liquidity, inflation, prices, and output? Third, what does speculative borrowing imply for productive investment and macroeconomic stability?

The contribution of the analysis is twofold. First, it introduces speculative activity directly into a monetary DSGE framework rather than treating it only indirectly through broad credit-demand relationships. Second, it provides a formal structure for studying how the allocation of bank credit between productive and speculative uses can affect money creation, inflationary pressure, and output dynamics.

Model Structure and Economic Mechanism

The model is a closed-economy cash-in-advance DSGE framework with three broad sectors: households, firms, and the government. Firms are divided into productive firms and a banking sector. Households own capital and labor, supply these inputs to firms, and allocate wealth across cash, demand deposits, time deposits, and bonds. Household utility depends on consumption and leisure. Because the model is cash-in-advance, consumption expenditure is constrained by available money balances and government transfers.

Productive firms operate competitively and use capital, labor, and bank loans as inputs. The key assumption is that loans formally extended for productive activity are not necessarily used entirely in production. A share of each loan enters production, while the remaining share may be diverted toward speculation when speculative returns are sufficiently attractive.

This allocation is governed by q, the productive loan-allocation parameter. A higher value of q means that a larger share of loans is used in production, while a lower value means that a larger share of borrowed funds may be diverted toward speculative activity. Since the productive use of credit is difficult to observe directly, the model is solved under four values of q= 0.2, 0.4, 0.6, and 0.8. These values are not interpreted as estimates of the true productive-credit share. They are used as a sensitivity exercise to examine whether the main mechanism remains present under alternative assumptions about the allocation of borrowed funds.

The banking sector receives demand and time deposits and extends loans to productive firms. The model includes reserve requirements on deposits and a multiplier-style relationship linking deposits, credit, and created money. This structure allows the model to trace how changes in speculative returns affect liquidity demand and money creation through the banking channel.

The government is modeled as a consolidated fiscal-monetary authority. It issues money, collects taxes, finances public expenditure, manages bonds, and supervises the banking system through reserve requirements. This treatment reflects the institutional motivation of the model, especially the limited independence of the monetary authority in the Iranian context.

The model is driven by two shocks. The first is a technology shock affecting productive capacity. The second is a speculative-return shock, which captures changes in the profitability of speculative activity across financial and non-financial markets. The main experiment studies the response of money, prices, inflation, and output to a positive shock to speculative returns.

The mechanism operates through the interaction between speculative returns and loan allocation. A positive speculative-return shock raises the attractiveness of diverting borrowed funds away from production. In the model, this increases liquidity demand and weakens the productive use of bank credit, generating higher money creation alongside weaker real activity.

Formal Model

The economy is populated by a representative household, productive firms, a banking sector, and a consolidated government-monetary authority. Time is discrete, and the economy is closed. The model follows a cash-in-advance structure in which consumption expenditure is constrained by available real money balances and transfers.

The representative household derives utility from consumption and leisure:

$$U_t = \sum_{t=0}^{\infty} \beta^t (C_t^{\eta} (1 - N_t)^{1-\eta})$$

where C_t denotes consumption, N_t labor supply, β the discount factor, and η the consumption weight in utility. The binding cash-in-advance constraint is written as:

$$c_t = m_t + \tau_t$$

where m_t denotes real money balances and τ_t government transfers. The household allocates wealth across capital, money balances, deposits, and bonds, while supplying labor and capital to firms. The household's intertemporal optimality conditions determine labor supply, consumption, money demand, bond holdings, and capital accumulation.

Capital evolves according to the standard accumulation equation:

$$K_{t+1} = K_t(1 - \delta\delta) + I_t$$

where $\delta\delta$ is the depreciation rate and I_t is investment. The notation $\delta\delta$ is used to distinguish capital depreciation from deposit-related parameters in the banking block.

The key mechanism of the model enters through the loan market. Productive firms borrow from the banking sector, but not all borrowed funds are necessarily used in production. Let $q \in (0,1)$ denote the share of bank loans allocated to productive activity. Then $l_t q$ enters production, while $l_t(1 - q)$ may be redirected toward speculative activity. A lower value of q therefore represents a larger speculative diversion of bank credit.

Productive output is given by:

$$Y_t = A_t\big(K_t(1 - z)\big)^{\alpha} \left(\frac{L_t}{P_t} qn\right)^{\gamma} \big(N_t(1 - x)\big)^{1-\alpha-\gamma}$$

where A_t is technology, K_t capital, $\frac{L_t}{P_t}$ real loans, P_t the price level, and x and z are the shares of labor and capital absorbed by the banking sector. The parameter γ measures the elasticity of output with respect to real loans.

Technology follows an autoregressive process:

$$lnA_t = \rho_A lnA_{t-1} + (1 - \rho_A) ln\bar{A} + \varepsilon_A$$

Speculative returns also follow an autoregressive process:

$$lni_{st} = \rho_{i_s} lni_{st-1} + \big(1 - \rho_{i_s}\big) ln\bar{i_s} + \varepsilon_{i_s}$$

where i_{st} is the return to speculative activity and ε_{i_s} is the speculative-return shock.

The banking sector receives demand and time deposits and extends loans to firms. A money-multiplier relationship links deposits, reserve requirements, and bank lending. In the model, created money is directly linked to bank lending through the following accounting relation:

$$CM_t = L_t = M_t(\frac{a}{\delta + rr_D} + \frac{b}{\mu + rr_T} - b - a)$$

where M_t is the monetary aggregate, a and b are the shares of money held as demand and time deposits, rr_D and rr_T are reserve requirements, and δ and μ denote the fractions of deposits with real equivalents.

A positive shock to ε_{i_s} raises the return to speculative activity and strengthens the incentive to divert borrowed funds away from production. If the banking sector accommodates the associated demand for liquidity, credit and money creation increase. Since part of borrowed funds no longer enters production, output declines, while the expansion of liquidity places upward pressure on prices and inflation.

The model is log-linearized around its calibrated steady state and simulated in Dynare. The full system includes the household optimality conditions, firm first-order conditions, banking-sector accounting relations, the government budget constraint, and the two exogenous shock processes. The full log-linearized system is omitted here for brevity but can be provided as supplementary material.

Calibration and Simulation

The model is calibrated using Iranian macroeconomic data and parameter values from previous DSGE and monetary-economics literature. Several monetary and banking parameters, including reserve ratios and the composition of money holdings across cash, demand deposits, and time deposits, are calculated from the Central Bank of the Islamic Republic of Iran's economic time-series data. Other structural parameters, including the discount factor, depreciation rate, labor-related parameters, capital share, technology persistence, and the elasticity of output with respect to loans, are taken from previous empirical and theoretical studies.

The discount factor, depreciation rate, labor-related parameters, and capital share are taken from Taghavi and Safarzadeh (2009). Technology persistence is calibrated using Manzoor and Taghipour (2016). The banking-sector labor-entry parameter follows Goodfriend and McCallum (2007), while the elasticity of output with respect to real loans follows Jensen, Kamath, and Bennett (1987). The speculative-return process is modeled as an autoregressive process, with persistence calibrated using information from speculative markets such as gold, foreign exchange, and equities, following Tripathy (2017), Erfani, Abounoori, and Safari (2017), and Moshiri and Seifi (2008).

A practical limitation is that the share of bank loans actually used for production is difficult to identify directly. For this reason, the model is solved under four values of q. This exercise is not intended to estimate the true productive share of loans. Instead, it provides a structured sensitivity analysis of the speculative-return mechanism under alternative assumptions about credit allocation. Since q enters the model as a fixed loan-allocation parameter in each simulation, changes in q may affect steady-state allocation more strongly than percentage impulse responses around the calibrated steady state. The exercise should therefore be interpreted as evidence on the direction and stability of the mechanism, not as an estimate of the exact empirical value of q.

The model is log-linearized and simulated in Dynare 4.6.4 and MATLAB R2017a. The main object of analysis is the impulse response of key macroeconomic variables to a one-percent positive shock to the speculative return.

Main Results

The impulse-response results indicate that positive shocks to speculative returns increase liquidity and money creation. Under the present log-linearized calibration, the simulations for q = 0.2, 0.4, and 0.6 generate numerically identical percentage responses. A one-percent speculative-return shock raises the model's monetary aggregate by approximately 2.0733 percent. Under the same shock, output falls by 0.1345 percent, while prices and inflation both rise by 1.1345 percent.

For q = 0.8, the direction of the response remains the same, although the magnitudes differ. A one-percent speculative-return shock increases the monetary aggregate by 2.0625 percent. Output falls by 0.1841 percent, while prices and inflation rise by 1.5525 percent. This comparison suggests that the qualitative direction of the mechanism is stable across the four q values, although the limited variation across the first three cases indicates that the current specification captures q mainly through calibration and steady-state structure rather than through rich dynamic sensitivity.

The results are summarized below.

Productive loan share q	**Money/liquidity response**	**Output response**	**Price response**	**Inflation response**
0.2	+2.0733%	-0.1345%	+1.1345%	+1.1345%
0.4	+2.0733%	-0.1345%	+1.1345%	+1.1345%
0.6	+2.0733%	-0.1345%	+1.1345%	+1.1345%
0.8	+2.0625%	-0.1841%	+1.5525%	+1.5525%

The identical responses for q = 0.2, 0.4, and 0.6 should be interpreted cautiously. They suggest that, within the current log-linearized specification, changes in q over this range mainly affect the allocation structure and steady-state interpretation of credit use rather than the percentage dynamic response to the speculative-return shock. I treat this as a limitation of the current calibration and sensitivity design. In ongoing work, I am examining whether alternative specifications of the loan-allocation mechanism, including making q endogenous or linking it more directly to speculative returns, generate richer variation across q values.

Overall, the simulations support the central hypothesis of the model: speculative-return shocks can be associated with monetary expansion and weaker output when bank credit is partly diverted away from productive use. The result is therefore both monetary and allocative: the shock increases liquidity and money creation while reducing the productive contribution of borrowed funds.

Policy Implications

The policy implications should be interpreted carefully. The model does not claim that taxation of speculative gains alone can solve inflation, excessive liquidity growth, or output weakness. Inflation in an economy such as Iran has multiple causes, including monetary, fiscal, institutional, and external factors. The model’s contribution is narrower: it identifies one possible channel through which speculative profitability may increase liquidity demand and contribute to money creation.

Within this framework, policies that reduce the private return to non-productive speculative activity may help moderate speculative credit demand and support the redirection of credit toward production. Taxation of speculative profits is one possible instrument, but the model should not be read as a full policy evaluation. Its implication is conditional: such measures are most relevant where speculative borrowing is large, financial supervision is weak, and banks accommodate liquidity demand. A complete policy analysis would require a richer treatment of taxation, enforcement, financial frictions, distributional effects, and substitution across speculative markets.

Concluding Remarks

This research note develops a closed-economy cash-in-advance DSGE model to examine whether speculative activity can contribute to bank money creation, inflationary pressure, and output weakness. The model is calibrated to the Iranian economy and includes a loan-allocation mechanism through which part of bank credit may be redirected away from production when speculative returns rise.

The results indicate that positive speculative-return shocks increase money creation, prices, and inflation while reducing output. These findings are consistent with a speculative money-supply channel, but they should be interpreted as model-based implications rather than as a complete empirical or policy evaluation.

Future work could extend the model in several directions. An open-economy version could incorporate exchange-rate dynamics and international financial linkages. The banking sector could be expanded to include multiple loan types, richer balance-sheet behavior, and stronger financial frictions. Given the importance of oil revenues in Iran, the government sector could be extended to include oil income and fiscal dependence on natural-resource revenues. Future research could also replace the cash-in-advance structure with a cash-credit framework or an overlapping-generations model to study speculative borrowing in alternative monetary environments.

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